CLASS 7 — MATHS · FORMULA SHEET

Chapter 1: **Large Numbers** | Quick revision, one page

Indian system

- Place values from the right: ones, tens, hundreds, thousands, ten thousands, lakhs, ten lakhs, crores, ten crores.
- Commas: first after three digits, then after every two digits, e.g. $12{,}34{,}56{,}789$.
- 1 lakh $= 1{,}00{,}000 = 10^5$. 1 crore $= 1{,}00{,}00{,}000 = 10^7 = 100$ lakh.

International system

- Commas after every three digits, e.g. $123{,}456{,}789$.
- 1 million $= 1{,}000{,}000 = 10^6 = 10$ lakh. 1 billion $= 10^9 = 1000$ million $= 100$ crore.
- 1 crore $= 10$ million.

Reading, writing and comparing

- Expanded form: $45{,}67{,}890 = 4 \times 10^6 + 5 \times 10^5 + 6 \times 10^4 + 7 \times 10^3 + 8 \times 10^2 + 9 \times 10$.
- More digits means a greater number. With equal digits, compare from the left, digit by digit.

Rounding and estimation

- Round to a place: look at the digit just to its right; 5 or more rounds up, less than 5 rounds down.
- Estimate before calculating: round each number and then add, subtract or multiply.

Operations

- Multiplying by $10, 100, 1000, \ldots$ adds that many zeros: $a \times 10^n$.
- Conversions: crore $\rightarrow$ lakh multiply by 100; lakh $\rightarrow$ thousand multiply by 100.

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